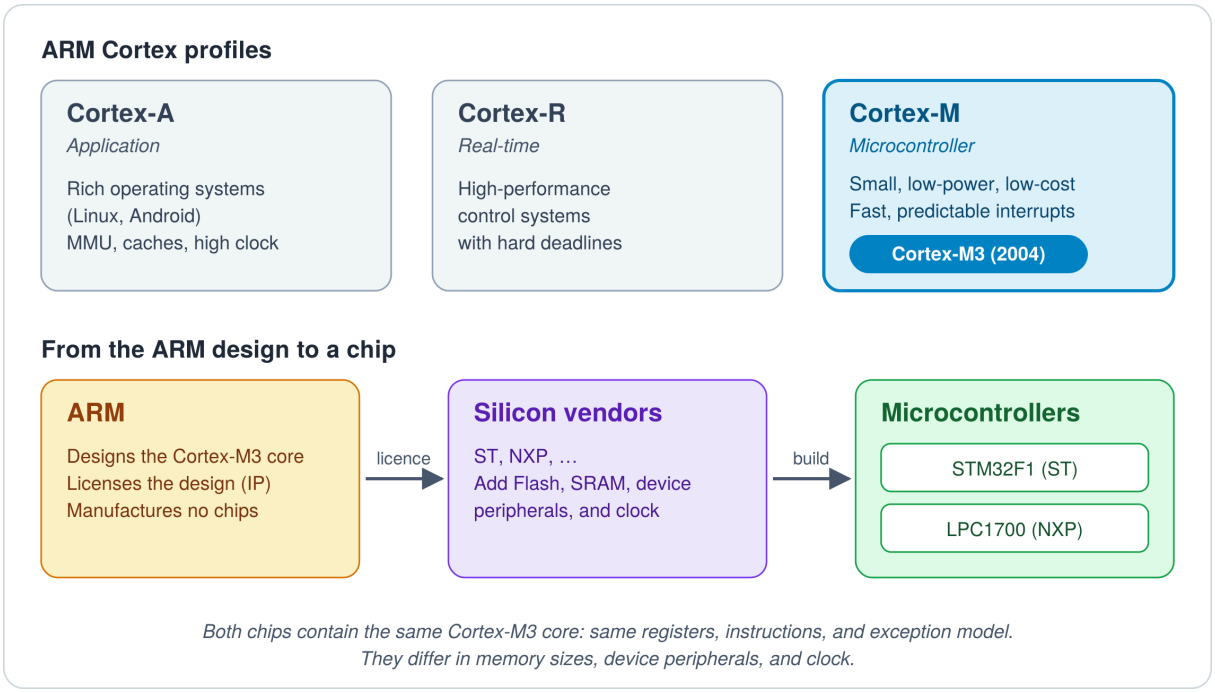
1 ARM Cortex-M3

The **Cortex-M3** is a 32-bit processor core designed by ARM for microcontrollers. Introduced in 2004, it belongs to the **M profile** of the ARM cores (*microcontroller*), which targets small, low-power, and low-cost systems with fast and predictable interrupt handling. The other two profiles are **A** (*application*), for processors that run operating systems such as Linux, and **R** (*real-time*), for high-performance control systems.



Both chips contain the same Cortex-M3 core: same registers, instructions, and exception model.
They differ in memory sizes, device peripherals, and clock.

ARM does not manufacture the chip: it licenses the design to silicon vendors, such as ST (STM32F1) and NXP (LPC1700), which integrate it with their own memories and peripherals.

Its main characteristics follow from the embedded constraints:

Feature	Cortex-M3
Architecture	ARMv7-M, 32-bit registers and addresses
Instruction set	Thumb-2 only, a mix of 16- and 32-bit instructions; hardware divide (`sdiv`, `udiv`)
Pipeline	3 stages: fetch, decode, execute
Buses	Separate instruction and data buses (Harvard-like), one shared 4 GB address space
Memory system	No cache, no MMU; optional MPU
Interrupts	Integrated interrupt controller (NVIC) with low, deterministic latency

1.1 Three Layers of Documentation

The documentation of a Cortex-M3 system is split into **three layers**, each defined by a different party. When looking for information, the first step is to know which layer answers the question.

- **Architecture (ARMv7-M)**: a **specification** written by ARM, not a physical chip. It fixes what software sees: the registers and their roles (the *programmer's model*), the instruction set, the way interrupts and faults are handled (the *exception model*), and the meaning of each region of the address space.
- **Core (Cortex-M3)**: the processor that ARM designs following that specification. It adds implementation details, such as the 3-stage pipeline, and a fixed set of *core peripherals*: the NVIC, the SysTick timer, the System Control Block, and an optional MPU.
- **Chip or board**: the physical product that a silicon vendor builds around the licensed core. The vendor decides how much Flash and SRAM exist, which device peripherals (UARTs, timers) are connected, and at which addresses.

Layer	Defined by	Examples of what it fixes
Architecture (ARMv7-M)	ARM	Registers, Thumb-2 instructions, exception numbers, address regions
Core (Cortex-M3)	ARM	NVIC, SysTick, SCB, MPU, 3-stage pipeline
Chip / board	Vendor	Memory sizes, device peripherals, IRQ assignment, clock

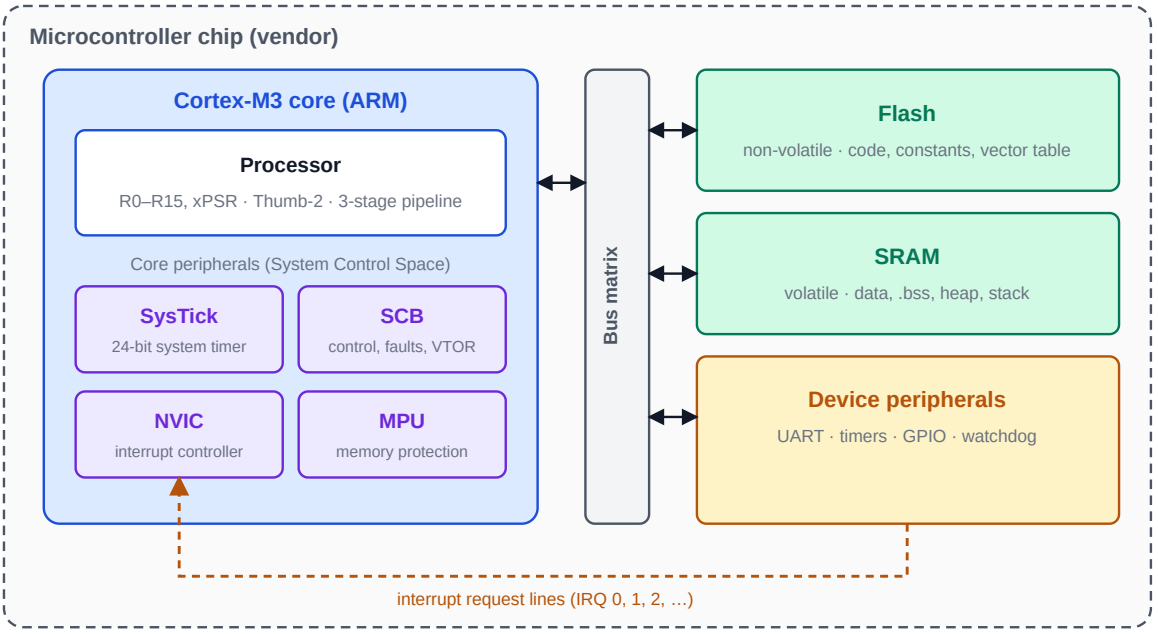


Figure 1.1: *Block diagram of a Cortex-M3 microcontroller: the ARM core with processor and core peripherals, the bus matrix, and the vendor's Flash, SRAM, and device peripherals.*

The chapters on the Cortex-M3 (programmer's model, memory map, execution modes and exceptions, core peripherals and MPU) cover the first two layers, which are identical on every Cortex-M3. Board-specific values are taken from the `mps2-an385` platform (an FPGA image of a Cortex-M3 system, emulated by QEMU, clocked at 25 MHz) and are marked as such.